

```

import pandas as pd

df=pd.read_excel(r"C:\Users\HP\Downloads\data.xlsx")
df

```

	CountryName	CountryCode	BirthRate	InternetUsers	\
0	Aruba	ABW	10.244	78.9	
1	Afghanistan	AFG	35.253	5.9	
2	Angola	AGO	45.985	19.1	
3	Albania	ALB	12.877	57.2	
4	United Arab Emirates	ARE	11.044	88.0	
..	...	...	...	...	
190	Yemen, Rep.	YEM	32.947	20.0	
191	South Africa	ZAF	20.850	46.5	
192	Congo, Dem. Rep.	COD	42.394	2.2	
193	Zambia	ZMB	40.471	15.4	
194	Zimbabwe	ZWE	35.715	18.5	

	IncomeGroup
0	High income
1	Low income
2	Upper middle income
3	Upper middle income
4	High income
..	...
190	Lower middle income
191	Upper middle income
192	Low income
193	Lower middle income
194	Low income


```

[195 rows x 5 columns]

id(df)

2443430181056

pd.__version__

'2.3.3'

len(df)

195

df.shape

(195, 5)

df.head()

```

```

      CountryName CountryCode BirthRate InternetUsers \
0           Aruba          ABW    10.244           78.9
1      Afghanistan          AFG    35.253           5.9
2           Angola          AGO    45.985          19.1
3          Albania          ALB    12.877          57.2
4  United Arab Emirates          ARE    11.044          88.0

      IncomeGroup
0      High income
1      Low income
2  Upper middle income
3  Upper middle income
4      High income

df.head(2)

      CountryName CountryCode BirthRate InternetUsers IncomeGroup
0           Aruba          ABW    10.244           78.9  High income
1  Afghanistan          AFG    35.253           5.9   Low income

df.tail()

      CountryName CountryCode BirthRate InternetUsers \
190      Yemen, Rep.          YEM    32.947           20.0
191      South Africa          ZAF    20.850           46.5
192  Congo, Dem. Rep.          COD    42.394           2.2
193           Zambia          ZMB    40.471           15.4
194          Zimbabwe          ZWE    35.715           18.5

      IncomeGroup
190  Lower middle income
191  Upper middle income
192      Low income
193  Lower middle income
194      Low income

df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 195 entries, 0 to 194
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  -
0   CountryName      195 non-null    object
1   CountryCode      195 non-null    object
2   BirthRate        195 non-null    float64
3   InternetUsers    195 non-null    float64
4   IncomeGroup      195 non-null    object
dtypes: float64(2), object(3)

```

```

memory usage: 7.7+ KB

df.columns

Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
      'IncomeGroup'],
      dtype='object')

len(df.columns)

5

df.tail()

      CountryName CountryCode BirthRate InternetUsers \
190      Yemen, Rep.         YEM      32.947          20.0
191      South Africa         ZAF      20.850          46.5
192  Congo, Dem. Rep.         COD      42.394           2.2
193          Zambia         ZMB      40.471          15.4
194        Zimbabwe         ZWE      35.715          18.5

      IncomeGroup
190  Lower middle income
191  Upper middle income
192      Low income
193  Lower middle income
194      Low income

df.dtypes

CountryName      object
CountryCode      object
BirthRate        float64
InternetUsers    float64
IncomeGroup      object
dtype: object

df

      CountryName CountryCode BirthRate InternetUsers \
0           Aruba         ABW      10.244          78.9
1    Afghanistan         AFG      35.253           5.9
2           Angola         AGO      45.985          19.1
3        Albania         ALB      12.877          57.2
4  United Arab Emirates         ARE      11.044          88.0
..          ...          ...          ...          ...
190      Yemen, Rep.         YEM      32.947          20.0
191      South Africa         ZAF      20.850          46.5
192  Congo, Dem. Rep.         COD      42.394           2.2
193          Zambia         ZMB      40.471          15.4
194        Zimbabwe         ZWE      35.715          18.5

```

```

            IncomeGroup
0            High income
1            Low income
2    Upper middle income
3    Upper middle income
4            High income
..            ...
190 Lower middle income
191 Upper middle income
192            Low income
193 Lower middle income
194            Low income

[195 rows x 5 columns]

df.columns
Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
      'IncomeGroup'],
      dtype='object')

df['CountryName']
0            Aruba
1    Afghanistan
2            Angola
3            Albania
4    United Arab Emirates
..            ...
190    Yemen, Rep.
191    South Africa
192    Congo, Dem. Rep.
193            Zambia
194    Zimbabwe
Name: CountryName, Length: 195, dtype: object

df['CountryCode']
0    ABW
1    AFG
2    AGO
3    ALB
4    ARE
..    ...
190  YEM
191  ZAF
192  COD
193  ZMB

```

```

194     ZWE
Name: CountryCode, Length: 195, dtype: object

df[['BirthRate', 'InternetUsers']]

   BirthRate  InternetUsers
0    10.244           78.9
1    35.253            5.9
2    45.985           19.1
3    12.877           57.2
4    11.044           88.0
..      ...           ...
190   32.947           20.0
191   20.850           46.5
192   42.394            2.2
193   40.471           15.4
194   35.715           18.5

[195 rows x 2 columns]

df_numerical_data = df[['BirthRate', 'InternetUsers']]
df_categorical_data = df[['CountryName', 'CountryCode', 'IncomeGroup']]
df_categorical_data.head(1)

   CountryName CountryCode  IncomeGroup
0      Aruba      ABW  High income

df_numerical_data.head(1)

   BirthRate  InternetUsers
0    10.244           78.9

print(df.shape)
print(df_numerical_data.shape)
print(df_categorical_data.shape)

(195, 5)
(195, 2)
(195, 3)

print(df.columns)
print('-----')
print(df_numerical_data.columns)

print(df_categorical_data.columns)
Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
      'IncomeGroup'],
      dtype='object')
-----

```

```

Index(['BirthRate', 'InternetUsers'], dtype='object')
Index(['CountryName', 'CountryCode', 'IncomeGroup'], dtype='object')

df[:]
```

	CountryName	CountryCode	BirthRate	InternetUsers	\
0	Aruba	ABW	10.244	78.9	
1	Afghanistan	AFG	35.253	5.9	
2	Angola	AGO	45.985	19.1	
3	Albania	ALB	12.877	57.2	
4	United Arab Emirates	ARE	11.044	88.0	
..	...	...	...	...	
190	Yemen, Rep.	YEM	32.947	20.0	
191	South Africa	ZAF	20.850	46.5	
192	Congo, Dem. Rep.	COD	42.394	2.2	
193	Zambia	ZMB	40.471	15.4	
194	Zimbabwe	ZWE	35.715	18.5	

```

IncomeGroup
0      High income
1      Low income
2  Upper middle income
3  Upper middle income
4      High income
..      ...
190 Lower middle income
191 Upper middle income
192      Low income
193 Lower middle income
194      Low income

[195 rows x 5 columns]

df[10:11]
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
10	Azerbaijan	AZE	18.3	58.7	Upper middle income

```

df[0:100:20]
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.90	High income
20	Belarus	BLR	12.500	54.17	Upper middle income
40	Costa Rica	CRI	15.022	45.96	Upper middle income
60	Gabon	GAB	30.555	9.20	Upper middle income
80	India	IND	20.291	15.10	Lower middle income

```

df[:,5]
```

	CountryName	CountryCode	BirthRate	InternetUsers	\
--	-------------	-------------	-----------	---------------	---

0	Aruba	ABW	10.244	78.900000
5	Argentina	ARG	17.716	59.900000
10	Azerbaijan	AZE	18.300	58.700000
15	Bangladesh	BGD	20.142	6.630000
20	Belarus	BLR	12.500	54.170000
25	Barbados	BRB	12.188	73.000000
30	Canada	CAN	10.900	85.800000
35	Cameroon	CMR	37.236	6.400000
40	Costa Rica	CRI	15.022	45.960000
45	Germany	DEU	8.500	84.170000
50	Ecuador	ECU	21.070	40.353684
55	Ethiopia	ETH	32.925	1.900000
60	Gabon	GAB	30.555	9.200000
65	Gambia, The	GMB	42.525	14.000000
70	Greenland	GRL	14.500	65.800000
75	Honduras	HND	21.593	17.800000
80	India	IND	20.291	15.100000
85	Israel	ISR	21.300	70.800000
90	Kazakhstan	KAZ	22.730	54.000000
95	Korea, Rep.	KOR	8.600	84.770000
100	Libya	LBY	21.425	16.500000
105	Lithuania	LTU	10.100	68.452900
110	Moldova	MDA	12.141	45.000000
115	Mali	MLI	44.138	3.500000
120	Mozambique	MOZ	39.705	5.400000
125	Namibia	NAM	29.937	13.900000
130	Netherlands	NLD	10.200	93.956400
135	Pakistan	PAK	29.582	10.900000
140	Poland	POL	9.600	62.849200
145	Qatar	QAT	11.940	85.300000
150	Sudan	SDN	33.477	22.700000
155	El Salvador	SLV	17.476	23.109300
160	Suriname	SUR	18.455	37.400000
165	Seychelles	SYC	18.600	50.400000
170	Tajikistan	TJK	30.792	16.000000
175	Tunisia	TUN	19.800	43.800000
180	Uruguay	URY	14.374	57.690000
185	Virgin Islands (U.S.)	VIR	10.700	45.300000
190	Yemen, Rep.	YEM	32.947	20.000000
IncomeGroup				
0	High income			
5	High income			
10	Upper middle income			
15	Lower middle income			
20	Upper middle income			

25 High income
 30 High income
 35 Lower middle income
 40 Upper middle income
 45 High income
 50 Upper middle income
 55 Low income
 60 Upper middle income
 65 Low income
 70 High income
 75 Lower middle income
 80 Lower middle income
 85 High income
 90 Upper middle income
 95 High income
 100 Upper middle income
 105 High income
 110 Lower middle income
 115 Low income
 120 Low income
 125 Upper middle income
 130 High income
 135 Lower middle income
 140 High income
 145 High income
 150 Lower middle income
 155 Lower middle income
 160 Upper middle income
 165 High income
 170 Lower middle income
 175 Upper middle income
 180 High income
 185 High income
 190 Lower middle income

df[:, -1]

	CountryName	CountryCode	BirthRate	InternetUsers	\
194	Zimbabwe	ZWE	35.715	18.5	
193	Zambia	ZMB	40.471	15.4	
192	Congo, Dem. Rep.	COD	42.394	2.2	
191	South Africa	ZAF	20.850	46.5	
190	Yemen, Rep.	YEM	32.947	20.0	
..	...	...	...	...	
4	United Arab Emirates	ARE	11.044	88.0	
3	Albania	ALB	12.877	57.2	
2	Angola	AGO	45.985	19.1	

1	Afghanistan	AFG	35.253	5.9
0	Aruba	ABW	10.244	78.9

	IncomeGroup
194	Low income
193	Lower middle income
192	Low income
191	Upper middle income
190	Lower middle income
..	...
4	High income
3	Upper middle income
2	Upper middle income
1	Low income
0	High income

[195 rows x 5 columns]

df[:, -5]

	CountryName	CountryCode	BirthRate	InternetUsers	\
194	Zimbabwe	ZWE	35.715	18.5000	
189	Samoa	WSM	26.172	15.3000	
184	Venezuela, RB	VEN	19.842	54.9000	
179	Ukraine	UKR	11.100	41.0000	
174	Trinidad and Tobago	TTO	14.590	63.8000	
169	Thailand	THA	11.041	28.9400	
164	Swaziland	SWZ	30.093	24.7000	
159	Sao Tome and Principe	STP	34.537	23.0000	
154	Sierra Leone	SLE	36.729	1.7000	
149	Saudi Arabia	SAU	20.576	60.5000	
144	French Polynesia	PYF	16.393	56.8000	
139	Papua New Guinea	PNG	28.899	6.5000	
134	Oman	OMN	20.419	66.4500	
129	Nicaragua	NIC	20.788	15.5000	
124	Malaysia	MYS	16.805	66.9700	
119	Mongolia	MNG	24.275	20.0000	
114	Macedonia, FYR	MKD	11.222	65.2400	
109	Morocco	MAR	21.023	56.0000	
104	Lesotho	LSO	28.738	5.0000	
99	Liberia	LBR	35.521	3.2000	
94	Kiribati	KIR	29.044	11.5000	
89	Japan	JPN	8.200	89.7100	
84	Iceland	ISL	13.400	96.5468	
79	Indonesia	IDN	20.297	14.9400	
74	Hong Kong SAR, China	HKG	7.900	74.2000	
69	Grenada	GRD	19.334	35.0000	

64	Guinea	GIN	37.337	1.6000
59	Micronesia, Fed. Sts.	FSM	23.511	27.8000
54	Estonia	EST	10.300	79.4000
49	Algeria	DZA	24.738	16.5000
44	Czech Republic	CZE	10.200	74.1104
39	Cabo Verde	CPV	21.625	37.5000
34	Cote d'Ivoire	CIV	37.320	8.4000
29	Central African Republic	CAF	34.076	3.5000
24	Brazil	BRA	14.931	51.0400
19	Bosnia and Herzegovina	BIH	9.062	57.7900
14	Burkina Faso	BFA	40.551	9.1000
9	Austria	AUT	9.400	80.6188
4	United Arab Emirates	ARE	11.044	88.0000

	IncomeGroup
194	Low income
189	Lower middle income
184	High income
179	Lower middle income
174	High income
169	Upper middle income
164	Lower middle income
159	Lower middle income
154	Low income
149	High income
144	High income
139	Lower middle income
134	High income
129	Lower middle income
124	Upper middle income
119	Upper middle income
114	Upper middle income
109	Lower middle income
104	Lower middle income
99	Low income
94	Lower middle income
89	High income
84	High income
79	Lower middle income
74	High income
69	Upper middle income
64	Low income
59	Lower middle income
54	High income
49	Upper middle income
44	High income

```

39  Lower middle income
34  Lower middle income
29      Low income
24  Upper middle income
19  Upper middle income
14      Low income
9      High income
4      High income

```

```
df.isnull()
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
..	...	...	...	...	...
190	False	False	False	False	False
191	False	False	False	False	False
192	False	False	False	False	False
193	False	False	False	False	False
194	False	False	False	False	False

```
[195 rows x 5 columns]
```

```
df.isna()
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
..	...	...	...	...	...
190	False	False	False	False	False
191	False	False	False	False	False
192	False	False	False	False	False
193	False	False	False	False	False
194	False	False	False	False	False

```
[195 rows x 5 columns]
```

```
df.isnull().sum()
```

```

CountryName    0
CountryCode    0
BirthRate      0
InternetUsers  0

```

```

IncomeGroup      0
dtype: int64

df.describe()

      BirthRate  InternetUsers
count  195.000000    195.000000
mean    21.469928    42.076471
std     10.605467    29.030788
min      7.900000     0.900000
25%     12.120500    14.520000
50%     19.680000    41.000000
75%     29.759500    66.225000
max     49.661000    96.546800

df_categorical_data.describe()

      CountryName  CountryCode  IncomeGroup
count           195           195           195
unique           195           195           4
top             Aruba           ABW  High income
freq              1              1           67

df_categorical_data.describe()

      CountryName  CountryCode  IncomeGroup
count           195           195           195
unique           195           195           4
top             Aruba           ABW  High income
freq              1              1           67

df.columns

Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
      'IncomeGroup'],
      dtype='object')

df.columns

Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
      'IncomeGroup'],
      dtype='object')

df.columns=['a','b','c','d','e']
df

      a      b      c      d      e
0      Aruba  ABW  10.244  78.9  High income
1  Afghanistan  AFG  35.253   5.9  Low income
2      Angola  AGO  45.985  19.1  Upper middle income
3      Albania  ALB  12.877  57.2  Upper middle income
4  United Arab Emirates  ARE  11.044  88.0  High income

```

```

..          ...    ...    ...    ...
190          Yemen, Rep.  YEM  32.947  20.0  Lower middle income
191          South Africa  ZAF  20.850  46.5  Upper middle income
192          Congo, Dem. Rep.  COD  42.394  2.2  Low income
193          Zambia  ZMB  40.471  15.4  Lower middle income
194          Zimbabwe  ZWE  35.715  18.5  Low income

```

```
[195 rows x 5 columns]
```

```
df.columns = ['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers', 'IncomeGroup']
df
```

```

          CountryName CountryCode BirthRate  InternetUsers \
0          Aruba          ABW      10.244           78.9
1    Afghanistan          AFG      35.253           5.9
2          Angola          AGO      45.985          19.1
3          Albania          ALB      12.877          57.2
4  United Arab Emirates          ARE      11.044          88.0
..          ...          ...          ...          ...
190          Yemen, Rep.          YEM      32.947          20.0
191          South Africa          ZAF      20.850          46.5
192          Congo, Dem. Rep.          COD      42.394           2.2
193          Zambia          ZMB      40.471          15.4
194          Zimbabwe          ZWE      35.715          18.5

```

```

          IncomeGroup
0          High income
1          Low income
2  Upper middle income
3  Upper middle income
4          High income
..          ...
190  Lower middle income
191  Upper middle income
192          Low income
193  Lower middle income
194          Low income

```

```
[195 rows x 5 columns]
```

```
df.columns
```

```

Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
      'IncomeGroup'],
      dtype='object')

```

```
df.BirthRate * df.InternetUsers
```

```
0      808.2516
```

```

1      207.9927
2      878.3135
3      736.5644
4      971.8720

```

```

...
190    658.9400
191    969.5250
192     93.2668
193    623.2534
194    660.7275

```

```
Length: 195, dtype: float64
```

```
df['myCalc'] = df.BirthRate * df.InternetUsers
```

```
df.head()
```

	CountryName	CountryCode	BirthRate	InternetUsers	\
0	Aruba	ABW	10.244	78.9	
1	Afghanistan	AFG	35.253	5.9	
2	Angola	AGO	45.985	19.1	
3	Albania	ALB	12.877	57.2	
4	United Arab Emirates	ARE	11.044	88.0	

	IncomeGroup	myCalc
0	High income	808.2516
1	Low income	207.9927
2	Upper middle income	878.3135
3	Upper middle income	736.5644
4	High income	971.8720

```
df = df.drop('myCalc',axis = 1)
```

```
df
```

	CountryName	CountryCode	BirthRate	InternetUsers	\
0	Aruba	ABW	10.244	78.9	
1	Afghanistan	AFG	35.253	5.9	
2	Angola	AGO	45.985	19.1	
3	Albania	ALB	12.877	57.2	
4	United Arab Emirates	ARE	11.044	88.0	
..	...	...	...	...	
190	Yemen, Rep.	YEM	32.947	20.0	
191	South Africa	ZAF	20.850	46.5	
192	Congo, Dem. Rep.	COD	42.394	2.2	
193	Zambia	ZMB	40.471	15.4	
194	Zimbabwe	ZWE	35.715	18.5	

	IncomeGroup
0	High income

```

1          Low income
2    Upper middle income
3    Upper middle income
4          High income
..          ...
190 Lower middle income
191 Upper middle income
192          Low income
193 Lower middle income
194          Low income

```

[195 rows x 5 columns]

df

	CountryName	CountryCode	BirthRate	InternetUsers	\
0	Aruba	ABW	10.244	78.9	
1	Afghanistan	AFG	35.253	5.9	
2	Angola	AGO	45.985	19.1	
3	Albania	ALB	12.877	57.2	
4	United Arab Emirates	ARE	11.044	88.0	
..	...	...	...	...	
190	Yemen, Rep.	YEM	32.947	20.0	
191	South Africa	ZAF	20.850	46.5	
192	Congo, Dem. Rep.	COD	42.394	2.2	
193	Zambia	ZMB	40.471	15.4	
194	Zimbabwe	ZWE	35.715	18.5	

```

IncomeGroup
0          High income
1          Low income
2    Upper middle income
3    Upper middle income
4          High income
..          ...
190 Lower middle income
191 Upper middle income
192          Low income
193 Lower middle income
194          Low income

```

[195 rows x 5 columns]

df['InternetUsers']

```

0      78.9
1       5.9
2      19.1

```

```

3      57.2
4      88.0
...
190    20.0
191    46.5
192     2.2
193    15.4
194    18.5
Name: InternetUsers, Length: 195, dtype: float64

```

```
df['InternetUsers'] < 2
```

```

0      False
1      False
2      False
3      False
4      False
...
190    False
191    False
192    False
193    False
194    False
Name: InternetUsers, Length: 195, dtype: bool

```

```
df[df['InternetUsers'] < 2]
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
11	Burundi	BDI	44.151	1.3	Low income
52	Eritrea	ERI	34.800	0.9	Low income
55	Ethiopia	ETH	32.925	1.9	Low income
64	Guinea	GIN	37.337	1.6	Low income
117	Myanmar	MMR	18.119	1.6	Lower middle income
127	Niger	NER	49.661	1.7	Low income
154	Sierra Leone	SLE	36.729	1.7	Low income
156	Somalia	SOM	43.891	1.5	Low income
172	Timor-Leste	TLS	35.755	1.1	Lower middle income

```
len(df[df['InternetUsers'] < 2])
```

```
9
```

```
df[df['BirthRate'] > 40]
```

	CountryName	CountryCode	BirthRate	InternetUsers	\
2	Angola	AGO	45.985	19.1	
11	Burundi	BDI	44.151	1.3	
14	Burkina Faso	BFA	40.551	9.1	
65	Gambia, The	GMB	42.525	14.0	
115	Mali	MLI	44.138	3.5	

127	Niger	NER	49.661	1.7
128	Nigeria	NGA	40.045	38.0
156	Somalia	SOM	43.891	1.5
167	Chad	TCD	45.745	2.3
178	Uganda	UGA	43.474	16.2
192	Congo, Dem. Rep.	COD	42.394	2.2
193	Zambia	ZMB	40.471	15.4

	IncomeGroup
2	Upper middle income
11	Low income
14	Low income
65	Low income
115	Low income
127	Low income
128	Lower middle income
156	Low income
167	Low income
178	Low income
192	Low income
193	Lower middle income

```
len(df[df['BirthRate'] > 40])
```

```
12
```

```
df[(df.BirthRate > 40) & (df.InternetUsers < 2)]
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
11	Burundi	BDI	44.151	1.3	Low income
127	Niger	NER	49.661	1.7	Low income
156	Somalia	SOM	43.891	1.5	Low income

```
df.BirthRate>40
```

0	False
1	False
2	True
3	False
4	False
	...
190	False
191	False
192	True
193	True
194	False

```
Name: BirthRate, Length: 195, dtype: bool
```

```
df[df.BirthRate>40]
```

	CountryName	CountryCode	BirthRate	InternetUsers	\
2	Angola	AGO	45.985	19.1	
11	Burundi	BDI	44.151	1.3	
14	Burkina Faso	BFA	40.551	9.1	
65	Gambia, The	GMB	42.525	14.0	
115	Mali	MLI	44.138	3.5	
127	Niger	NER	49.661	1.7	
128	Nigeria	NGA	40.045	38.0	
156	Somalia	SOM	43.891	1.5	
167	Chad	TCD	45.745	2.3	
178	Uganda	UGA	43.474	16.2	
192	Congo, Dem. Rep.	COD	42.394	2.2	
193	Zambia	ZMB	40.471	15.4	

	IncomeGroup
2	Upper middle income
11	Low income
14	Low income
65	Low income
115	Low income
127	Low income
128	Lower middle income
156	Low income
167	Low income
178	Low income
192	Low income
193	Lower middle income

```
Filter = df.InternetUsers < 2
```

```
Filter2 = df.BirthRate >40
```

```
df[Filter & Filter2]
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
11	Burundi	BDI	44.151	1.3	Low income
127	Niger	NER	49.661	1.7	Low income
156	Somalia	SOM	43.891	1.5	Low income

```
df['IncomeGroup']
```

0	High income
1	Low income
2	Upper middle income
3	Upper middle income
4	High income
	...
190	Lower middle income
191	Upper middle income

```

192         Low income
193     Lower middle income
194         Low income
Name: IncomeGroup, Length: 195, dtype: object

df['IncomeGroup'].unique()

array(['High income', 'Low income', 'Upper middle income',
      'Lower middle income'], dtype=object)

df['IncomeGroup'].nunique()

4

df[df.IncomeGroup == 'high income']

Empty DataFrame
Columns: [CountryName, CountryCode, BirthRate, InternetUsers, IncomeGroup]
Index: []

df[df.IncomeGroup == 'Low income']

```

	CountryName	CountryCode	BirthRate	InternetUsers	\
1	Afghanistan	AFG	35.253	5.90	
11	Burundi	BDI	44.151	1.30	
13	Benin	BEN	36.440	4.90	
14	Burkina Faso	BFA	40.551	9.10	
29	Central African Republic	CAF	34.076	3.50	
38	Comoros	COM	34.326	6.50	
52	Eritrea	ERI	34.800	0.90	
55	Ethiopia	ETH	32.925	1.90	
64	Guinea	GIN	37.337	1.60	
65	Gambia, The	GMB	42.525	14.00	
66	Guinea-Bissau	GNB	37.503	3.10	
77	Haiti	HTI	25.345	10.60	
93	Cambodia	KHM	24.462	6.80	
99	Liberia	LBR	35.521	3.20	
111	Madagascar	MDG	34.686	3.00	
115	Mali	MLI	44.138	3.50	
120	Mozambique	MOZ	39.705	5.40	
123	Malawi	MWI	39.459	5.05	
127	Niger	NER	49.661	1.70	
132	Nepal	NPL	20.923	13.30	
148	Rwanda	RWA	32.689	9.00	
154	Sierra Leone	SLE	36.729	1.70	
156	Somalia	SOM	43.891	1.50	
158	South Sudan	SSD	37.126	14.10	
167	Chad	TCD	45.745	2.30	
168	Togo	TGO	36.080	4.50	
177	Tanzania	TZA	39.518	4.40	

178	Uganda	UGA	43.474	16.20
192	Congo, Dem. Rep.	COD	42.394	2.20
194	Zimbabwe	ZWE	35.715	18.50

	IncomeGroup
1	Low income
11	Low income
13	Low income
14	Low income
29	Low income
38	Low income
52	Low income
55	Low income
64	Low income
65	Low income
66	Low income
77	Low income
93	Low income
99	Low income
111	Low income
115	Low income
120	Low income
123	Low income
127	Low income
132	Low income
148	Low income
154	Low income
156	Low income
158	Low income
167	Low income
168	Low income
177	Low income
178	Low income
192	Low income
194	Low income

```
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
plt.rcParams['figure.figsize'] = 6,2 # rcParams -- Reduce the size-6 width and
import warnings
warnings.filterwarnings('ignore') # dont display/ignore the warning messages
#pandas- Dataframes
#Numpy -- nd array
#Matplotlib--Visualizaion
#Seaborn- Distribution Visualization
```

```
df['InternetUsers']
```

```
0    78.9  
1     5.9  
2    19.1  
3    57.2  
4    88.0
```

```
...
```

```
190   20.0
```

```
191   46.5
```

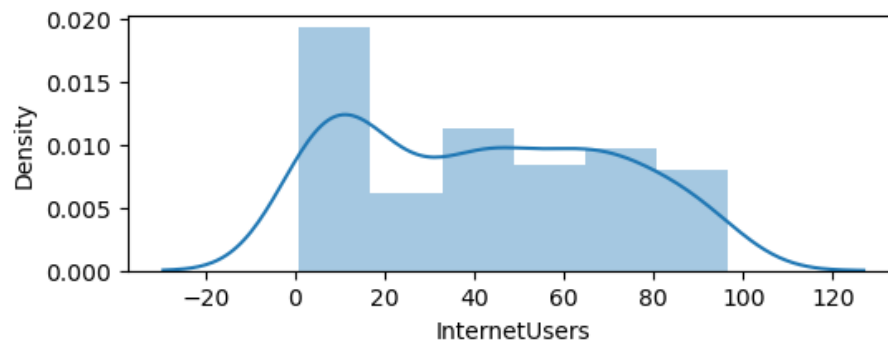
```
192    2.2
```

```
193   15.4
```

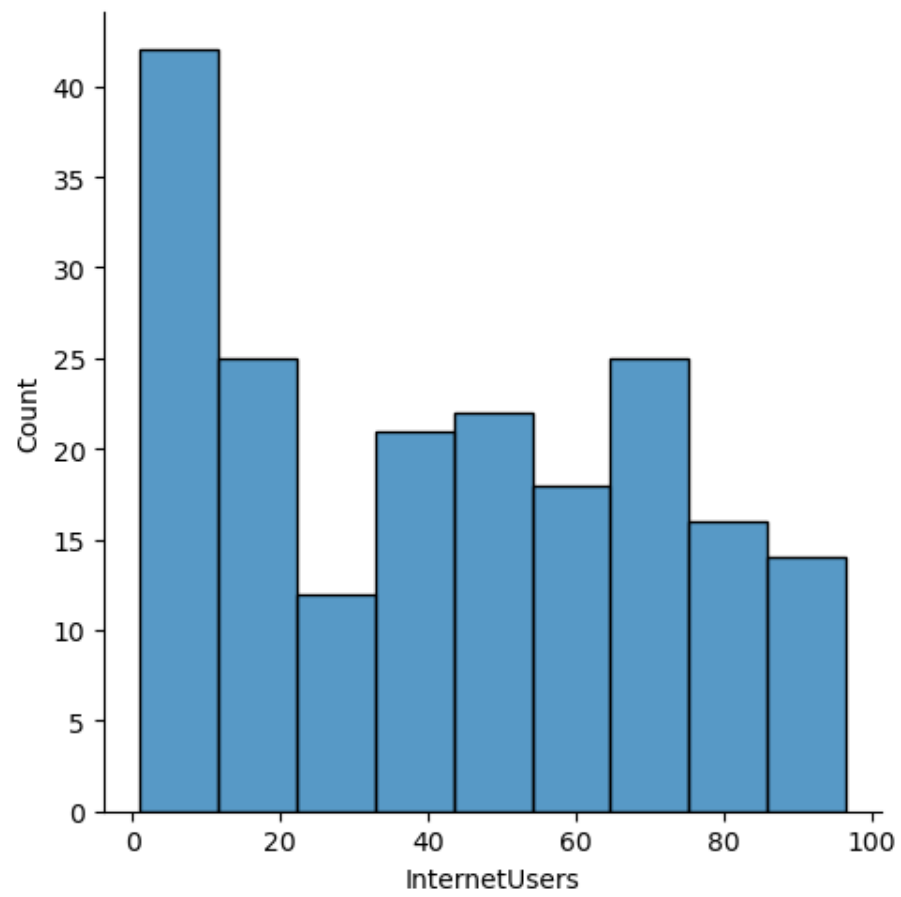
```
194   18.5
```

```
Name: InternetUsers, Length: 195, dtype: float64
```

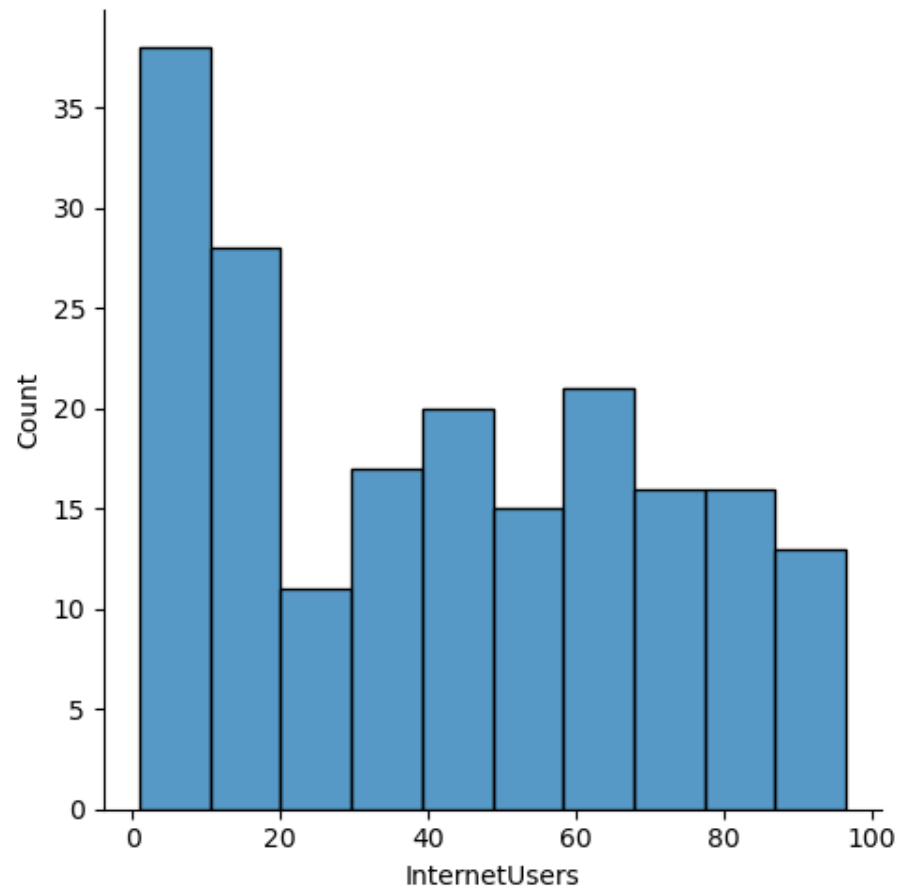
```
vis1 = sns.distplot(df['InternetUsers'])
```



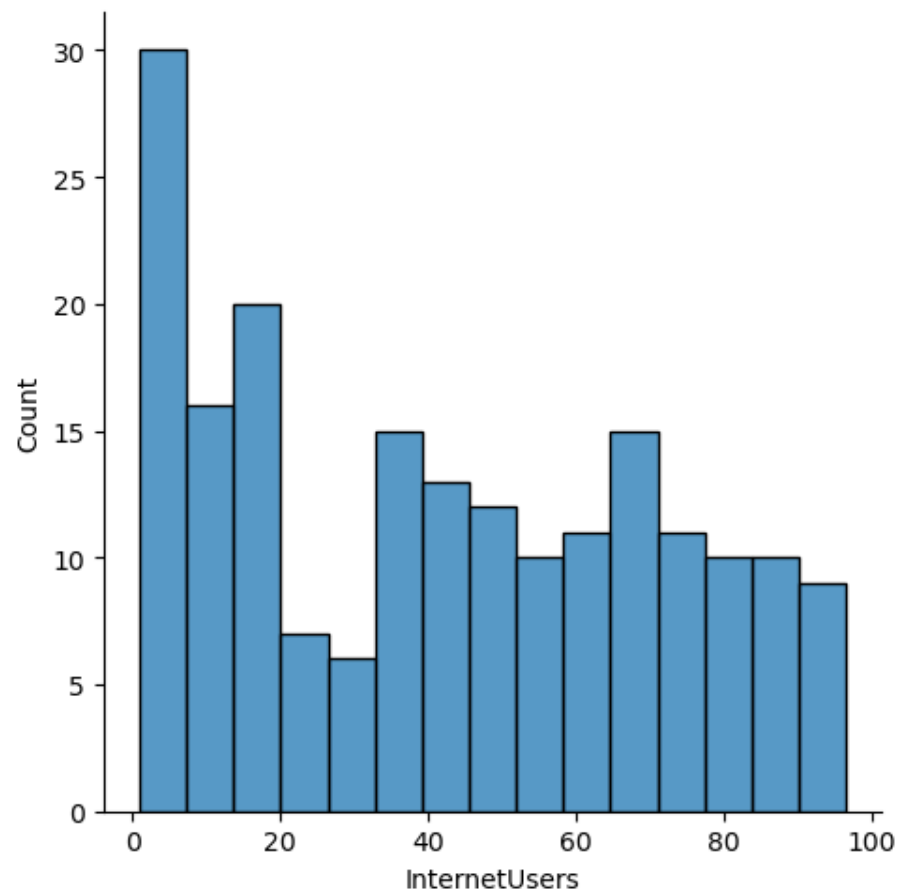
```
vis2 = sns.displot(df['InternetUsers'])
```



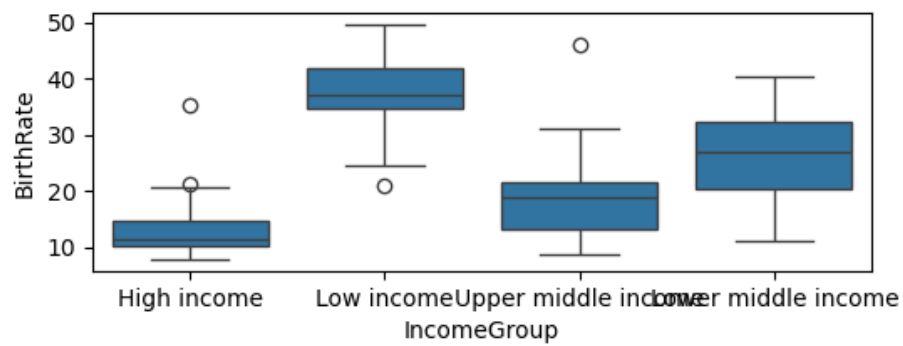
```
vis2 = sns.displot(df['InternetUsers'] , bins=10)
```



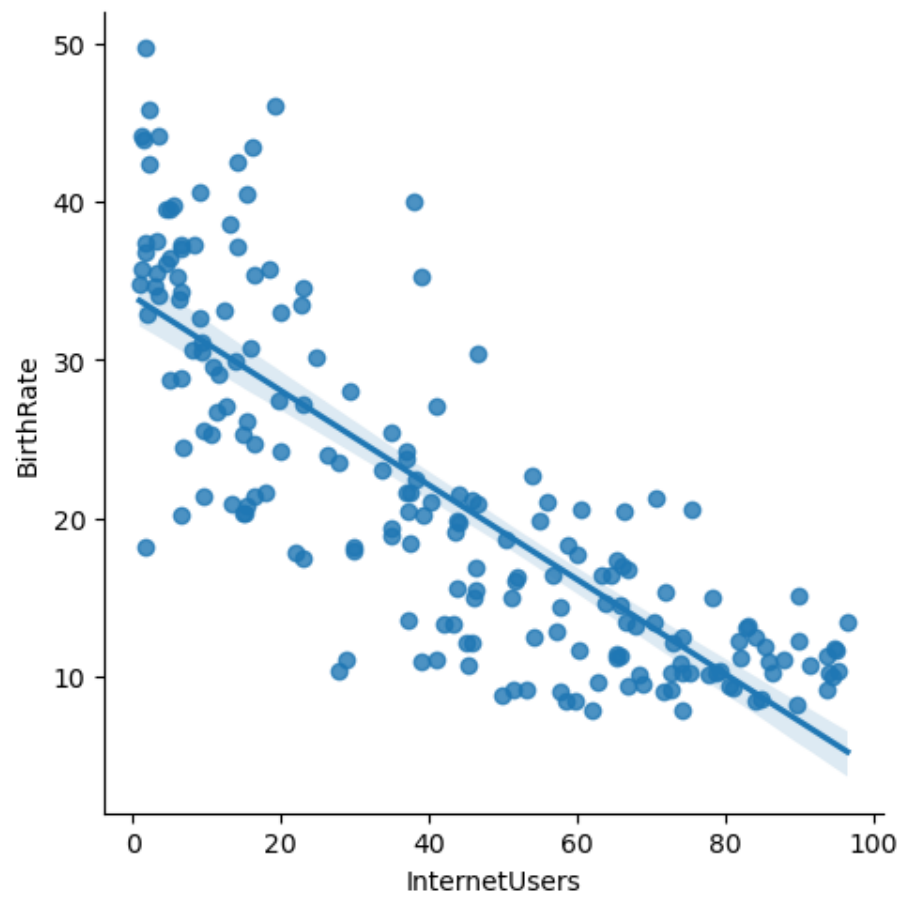
```
vis2 = sns.displot(df['InternetUsers'] , bins=15)
```



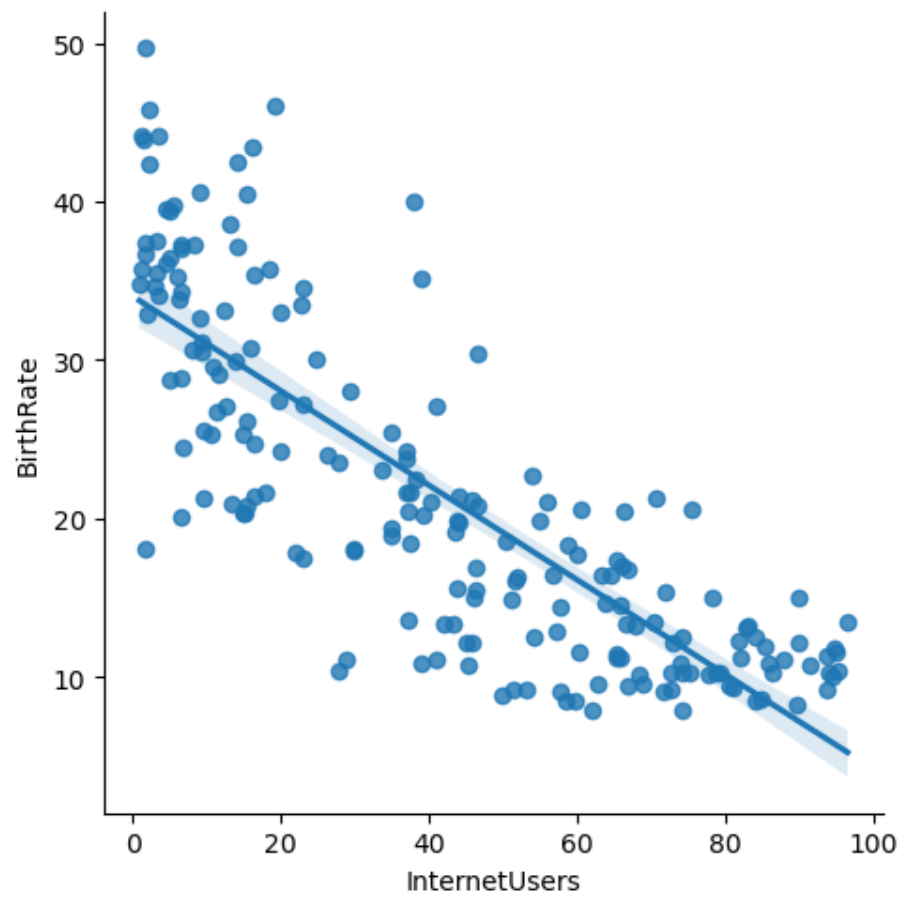
```
vis3 = sns.boxplot(data = df, x='IncomeGroup' , y='BirthRate')
```



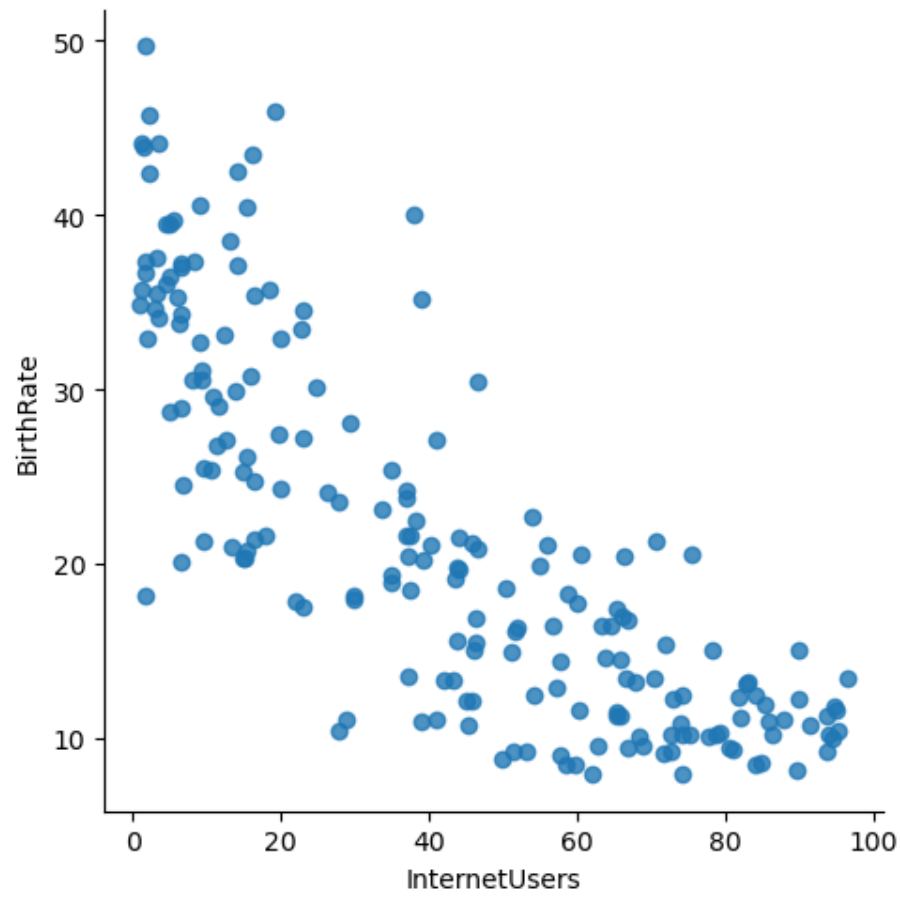
```
vis5 = sns.lmplot(data = df, x = 'InternetUsers' , y = 'BirthRate')
```

```
vis5 = sns.lmplot(data = df, x = 'InternetUsers' , y = 'BirthRate', fit_reg= True)
```



```
vis6 = sns.lmplot(data = df, x = 'InternetUsers' , y = 'BirthRate', fit_reg= False)
```



```
vis6 = sns.lmplot(data = df, x = 'InternetUsers' , y = 'BirthRate', fit_reg= False)
```

